Report

Current Biology

Arm Coordination in Octopus Crawling Involves Unique Motor Control Strategies

Highlights

- Octopuses use unique strategies to coordinate their arms in crawling
- They can crawl in any direction relative to the facing (body) direction
- Arms create pushing thrust by simple shortening and then elongating movements
- Arm coordination in octopus crawling lacks any obvious rhythmical pattern

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In Brief

Levy et al. show that octopuses use non-rhythmical arm coordination for crawling in any direction, while at the same time independently controlling their body orientation. They are able to do so because they decide in a moment-to-moment fashion which of the eight arms to recruit for pushing the body with simple shortening and then elongating movements.

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Report

Arm Coordination in Octopus Crawling Involves Unique Motor Control Strategies

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SUMMARY

To cope with the exceptional computational complexity that is involved in the control of its hyper-redundant arms [1], the octopus has adopted unique motor control strategies in which the central brain activates rather autonomous motor programs in the elaborated peripheral nervous system of the arms [2, 3]. How octopuses coordinate their eight long and flexible arms in locomotion is still unknown. Here, we present the first detailed kinematic analysis of octopus arm coordination in crawling. The results are surprising in several respects: (1) despite its bilaterally symmetrical body, the octopus can crawl in any direction relative to its body orientation; (2) body and crawling orientation are monotonically and independently controlled; and (3) contrasting known animal locomotion, octopus crawling lacks any apparent rhythmical patterns in limb coordination, suggesting a unique non-rhythmical output of the octopus central controller. We show that this uncommon maneuverability is derived from the radial symmetry of the arms around the body and the simple pushing-by-elongation mechanism by which the arms create the crawling thrust. These two together enable a mechanism whereby the central controller chooses in a moment-to-moment fashion which arms to recruit for pushing the body in an instantaneous direction. Our findings suggest that the soft molluscan body has affected in an embodied way [4, 5] the emergence of the adaptive motor behavior of the octopus.

RESULTS

Octopus vulgaris, a member of the bilaterian clade, has a bilaterally symmetrical body plan [6]. Its two eyes and optic lobes are set on the sides of the head, and they are connected to the central brain, which is situated inside a cartilaginous capsule in the center of the head. The mouth of the octopus has a fixed position relative to the brain and thus, also to the head and eye sockets. As in all modern cephalopods (Coleoidea), the arms emerge close to the mouth in a fixed position and orientation around it (see Figure S1B). The arms can be seen as having a bilaterally symmetrical organization (four on each side) but also as having radial symmetry, as they emerge in a circle around the mouth with a 45° angle between adjacent arms. The morphology of the head [7] prevents its rotation relative to the arms, causing body orientation and facing direction to be the same. The octopus thus differs from animals that have a flexible neck allowing separation between the head and body orientation systems. Therefore, we used the line running between the bases of the four left arms (L1–L4) and the four right arms (R1–R4) as an accurate measurement of the octopus's body and head orientation (the mantle lies on the rear side, Figure S1B; see Supplemental Experimental Procedures). We used positive and negative ranges of 0° to ±180° to describe, respectively, crawling into the right or left hemi-plane relative to the longitudinal axis of the body (see inset in Figure 3). All experiments were carried out according to the regulations of the Hebrew University of Jerusalem (see Supplemental Experimental Procedures for details).

Data from 145 non-consecutive, and thus independent, pairs of video frames showed that octopuses virtually crawled in all directions relative to their body direction with a mean value of approximately zero and a very large variance (mean ± SD was $-4.8° \pm 88.3°$). Negative and positive crawling direction values (to the left or right hemi-plane, respectively) were symmetrically distributed, as the median value was almost precisely zero, and the two-sample Kolmogorov-Smirnov test was also unable to reject the hypothesis of symmetrical distribution ($p \approx 0.4$). The direction was not normally distributed around zero (Lilliefors test, $p < 0.01$), but the absolute value of crawling direction (i.e., between 0° and 180°; Figure 1C) was a skewed Γ (gamma) distribution around a mode value of about 45° (Kolmogorov-Smirnov test $k \approx 2.13$ and $\theta \approx 35.68$, $p > 0.2$; red line in Figure 1C). This shows that despite the very large variation in crawling direction, overall octopuses preferred crawling in an average direction of ±45° relative to their body orientation.

During crawling, octopuses could monotonically change their crawling direction while maintaining a fixed body orientation (Figure 1A; Movie S1) and vice versa (Figure 1B; Movie S2), suggesting that these two parameters are controlled independently. To test this, we collected data from 100 non-consecutive video frames, and body rotations and changes in crawling direction

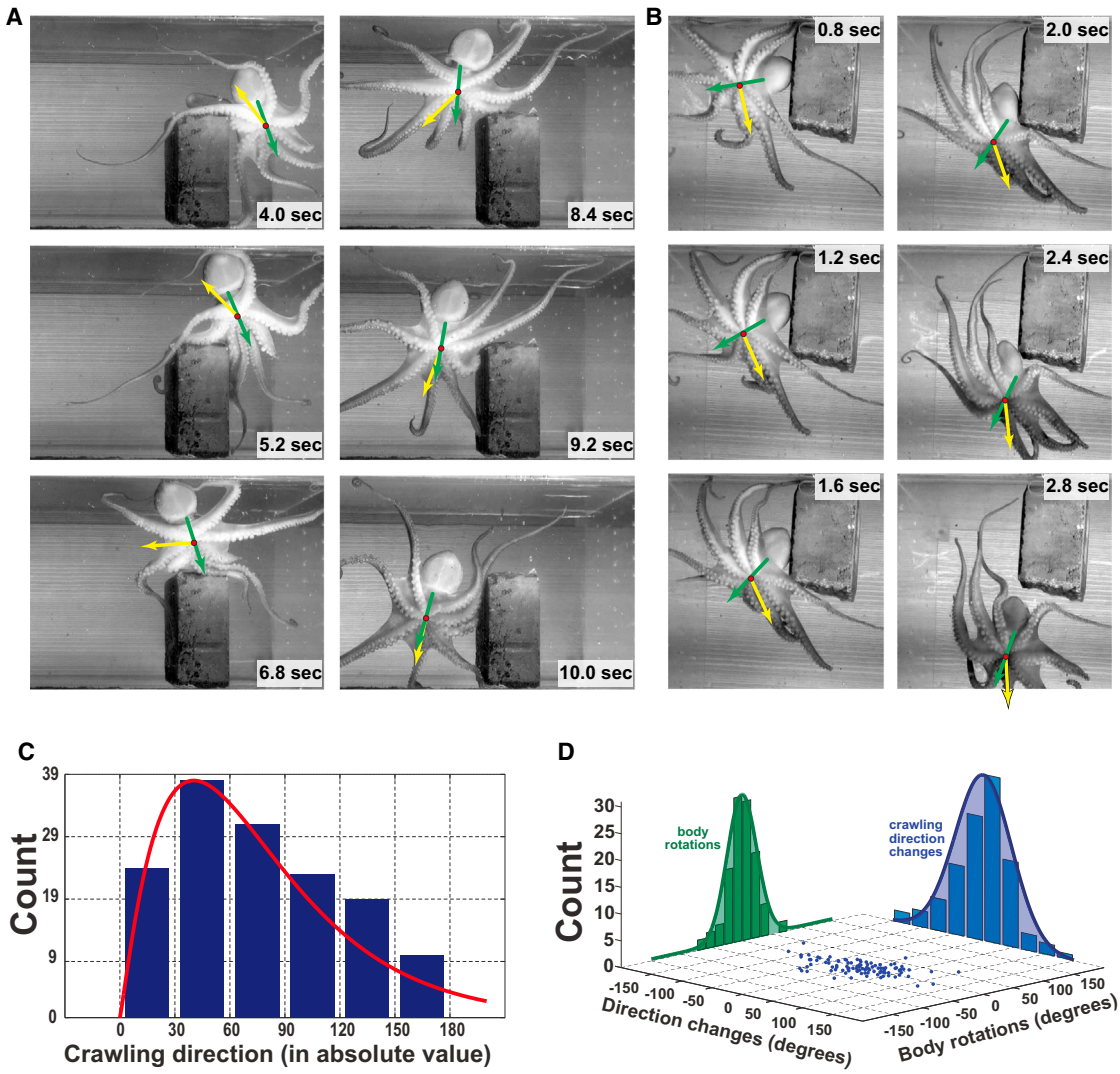


Figure 1. Octopuses Crawl in All Directions Relative to Body Direction

(A and B) Six time frames from two crawling sequences. Green arrows mark body direction; yellow arrows give the crawling direction.

(C) Histogram showing the absolute values of crawling direction from 145 pairs of time frames with the fitted Γ (gamma) distribution (red line).

(D) Histograms of body rotations (green) and changes in crawling direction (blue) from 100 time frames with the fitted normal distributions. Changes in body direction as a function of changes in crawling direction are shown on the XY plane (blue dots).

See also Movies S1 and S2.

were calculated over 2.8-s periods. Both parameters were found to be normally distributed around approximately zero (Lilliefors test, $p > 0.2$ for both; Figure 1D). However, Pearson's linear correlation test showed no correlation between them ($\rho \approx 0.12$, $p \approx 0.23$), confirming their independence. Body direction (and therefore also facing direction) appeared more stable than crawling direction with respect to the external world (Figure 1D; SD with 95% confidence intervals was 21.39°–28.30° for rotations and 40.87°–50.08° for direction changes).

The biomechanics of crawling is based on stereotypical arm shortening and then elongating movements. This stereotypical arm behavior that pushes the body (described in detail in the Supplemental Experimental Procedures) involves shortening of a proximal arm segment, followed by a group of suckers anchoring to the substrate, and then elongation of the proximal segment, thereby generating the pushing thrust to move the body. This movement resembles locomotion patterns in other flexible worm-like animals [8, 9], including other mollusks [9–11]. The independent control of body rotation was achieved by one of two control strategies, both based on the fact that the angle between the base of each arm and the body remained constant, and both are described qualitatively in the Supplemental Experimental Procedures.

We next wished to understand the mechanism enabling the octopus to change its crawling direction smoothly. If arms do indeed push the body mainly by elongation, and if arms pushing simultaneously apply equal forces, then it should be possible to calculate the direction of crawling using the geometrical fact that

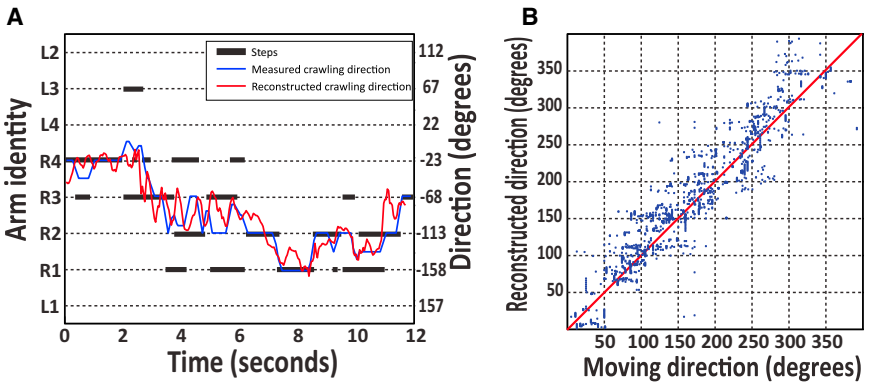


Figure 2. Arms Push the Body Only by Elongating and Apply Equal Pushing Forces when Working Together

(A) Stepping record of a ~12-s movement. Arm identities are given on the left and their fixed pushing direction on the right. The measured (blue) and predicted (red) crawling directions in body orientation coordinates are superimposed.

(B) Reconstructed crawling direction (y axis) and measured crawling direction (x axis) taken from 1,636 time frames (Pearson linear correlation test values: $\rho \approx 0.94$, $p < 0.001$).

See also Figure S2.

each arm has a fixed pushing direction, which is dictated by its position on the body (see inset in Figures 3 and S1B). Thus, the predicted crawling direction was calculated as the simple (weightless) vectorial summation of the pushing arms. Figure 2A shows the stepping record of a continuous crawl of about 12 s, with the time intervals during which each arm actively participated in the pushing. The identity of each arm is given on the left axis, and its predefined pushing direction is also shown in the same graph (right axis). The superposition of the predicted (red) and measured (blue) crawling direction shows the high level of precision of this reconstructed crawling direction. This confirmed our assumption that the arms push only by elongating and arms that push together apply equal forces. The direction of crawling is therefore determined simply by the identity of the pushing arms. The highly significant correlation (Figure 2B) between the reconstructed and the measured crawling direction from 1,636 time frames (Pearson's linear correlation test, $\rho \approx 0.94$, $p < 0.001$) conclusively supports our hypothesis.

Figure 3A demonstrates that octopuses generally prefer to use some arms over others in crawling (the test for significance of these preferences is detailed in Supplemental Experimental Procedures). More specifically, octopuses showed preference for pushing with the four hind arms and a rather small but significant preference for using the third arm over the fourth on each side. Measuring the frequency of using specific pairs of arms (Figure 3B and statistical analysis in Supplemental Experimental Procedures) also revealed a general preference for using specific pairs of arms, particularly for pushing with the third arm together with the fourth arm on the same side of the body or the second arm on one side together with the fourth arm on the other side. Both types of coupling arm pairs fit the distribution of crawling direction around ±45° (Figure 1C), providing additional and independent support for the idea that the identity of the active arms is what determines the crawling direction. This idea is further supported by the finding that L4 + R4 activation, which would result in forward crawling, was less significant than the combinations that would result in the preferred ±45° crawling direction (Figure 3, and see statistical analysis in Supplemental Experimental Procedures).

We next analyzed the pattern of arm recruitment in crawling. Inspection of octopus stepping records like those in Figures 2A and 4A1–4A4 suggested that although the arms may have some independent rhythmical repetitive patterns (e.g., traces of arms L2, L3, and R4 in Figure 4A1 and arms L4 and R3 in Figure 4A3), the coordination between the arms lacked any obvious rhythmical pattern. This lack of a clear rhythmical pattern is especially highlighted when compared with examples of the stepping records of locomotion in other animals, such as walking in stick insects and in *Drosophila* (Figures S3A2 and S3A1, based on [12] and [13], respectively). Intrinsic rhythmical repetitive patterns of locomotion can be easily detected in the records of the insect stepping movements. This contrasts sharply to the four examples of stepping records of the octopus crawling movements in Figures 4A1–4A4, which show what seems to be "erratic," unorganized stepping.

To look for possible hidden rhythmicities in arm coordination in octopus crawling, we analyzed the instantaneous crawling velocity (i.e., mouth velocity) (superimposed on the stepping records in Figures 4A1–4A4, blue) and used fast Fourier transform (FFT) to extract power spectra of the frequencies from each of these velocity profiles (Figures 4B1–4B4). No dominant frequencies could be found in any of the velocity profiles; the FFT analysis showed several dominant frequencies that merely reflected the frequencies related to the time of the entire movement (or one-half or one-third of it). No clear pattern was found in arm recruitment even when octopuses crawled on a straight path (e.g., Figures 4A1, 4A2, 4B1, and 4B2), suggesting that the erratic order is not associated with the changing of crawling direction. Applying a similar FFT technique to the velocity profiles of the walking stick insect and *Drosophila* extracted a single clear and dominant frequency for each (Figure S3). This confirmed both that in each of these species, a single dominant frequency is present in its velocity profile and that it could be extracted by FFT.

In further searching for an organized temporal pattern in octopus crawling, we collected the step durations from 12 continuous movements and normalized them to the average velocity of the movement. Then we examined the output sequence using the extended distribution free version of the Wald-Wolfowitz runs test [14], described in [15] (details about this test are given in Supplemental Experimental Procedures). The sequence was found to be random (number of runs: 48 $z = -1.0348$, $p = 0.3008$). Applying the same procedure to the data from the *Drosophila* and stick insect showed lack of randomness in their stepping (see Supplemental Experimental Procedures).

Therefore, failure of the FFT to extract dominant frequencies from octopus crawling records together with the randomness of step duration suggest that arm coordination in octopus crawling may involve mechanisms that are different from the

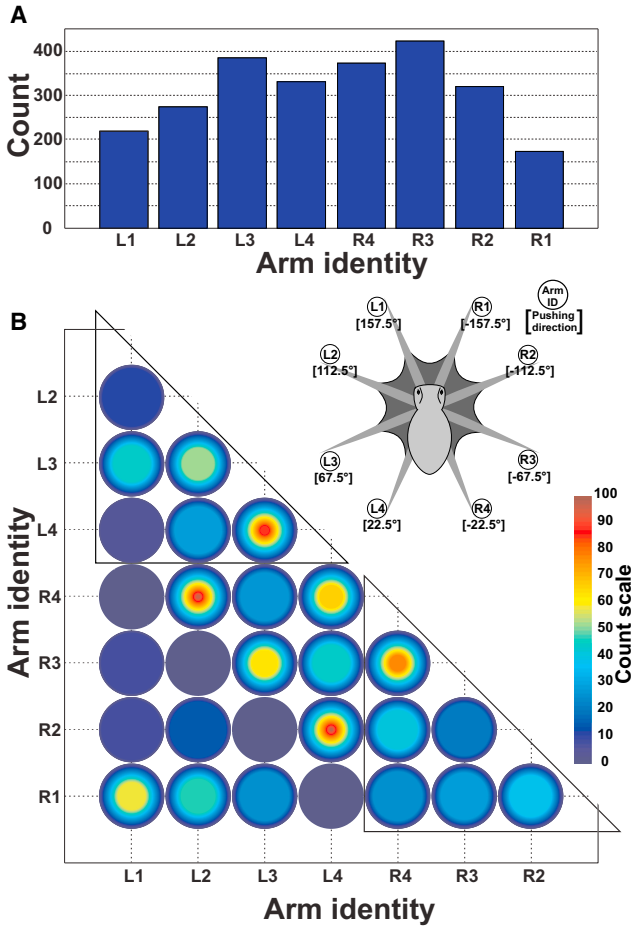


Figure 3. Distribution of Arm Usage during Crawling

(A) Histogram showing the distribution of arm usage in pushing (2,506 time frames). Arm identity is given below each bar.

(B) Graphical pseudo-3D (truncated cone from above) depiction of the distribution of pairs of arms used together in pushing (980 time frames). The frequency color scale is given on the right (from blue up to red). The upper left triangle marks pairs of arms from the left side of the body, the lower right triangle marks pairs of arms from the right side of the body, and the remainders are pairs where one arm is from the right side and the other is from the left side of the body. Note that the pairs used most frequently together give a crawling direction of 45°.

The inset identifies the arms and gives their pushing direction relative to the body.

See also Figure S1.

conventional central pattern generator (CPG)-driven rhythmicities involved in locomotion of other animals [16–18]. The runs test rejected the hypothesis of randomness in the spatial pattern of multiple arm recruitment during octopus crawling, showing that not all aspects in the crawling are random, as would be expected since the animal succeeds in crawling in a non-random (desired) direction after all.

DISCUSSION

The flexible body and arms of the octopus are bilaterally symmetrical, but the arms are also distributed with a fixed radial symmetry around the mouth. Radially organized animals (e.g., adult starfish and brittle stars) can crawl in any direction relative to their body orientation [19, 20]. The radial distribution of the arms provides the octopus with the same physical ability. Yet, what we have found here shows that unlike the brittle star and like other animals with bilateral body organization, the octopus still coordinates crawling direction with its body orientation in order to control the direction of its sensing and feeding apparatuses relative to the environment.

Several aspects of octopus vision and body plan may explain why the preferred average crawling direction is ±45° relative to its body orientation (Figure 1C). The octopus eye lacks a fovea centralis, having instead a central horizontal strip which may emphasize vision along the horizontal axis of each eye [21]. Therefore, despite the absence of head rotations and only limited eye movements relative to the body [7, 21, 22], the horizontal field of vision of each eye covers roughly 180° with virtually zero intersection. We speculate that octopuses prefer not to crawl forward or backward (0° or 180°, respectively) relative to the body direction because this would place the target at the border line of the visual fields of the two eyes. The preferred average crawling direction of the octopus (±45°) fits the generalization that most animals with bilateral symmetry move in the direction (not necessarily parallel to body orientation) in which the sensory systems can function effectively [6]. However, in the octopus, the deviation around this preferred direction of locomotion is very large. As explained above, this may be due to the fact that the field of vision is largely independent of body orientation. This relaxes the tight relationship between vision and the direction of locomotion, enabling the octopus to change its body orientation (Figure 1D) and still follow a target.

The stereotypical pushing-by-elongation mechanism simplifies the control of crawling because as in goal-directed arm movements, it can be generated by a simple peripheral motor program, thus freeing the central control from the complex task of directly controlling arm behavior [4, 23]. Elongation in muscular-hydrostatic structures like the octopus arm [24, 25] may be achieved by a rather simple command of a global simultaneous activation of the transverse muscles relative to the longitudinal muscles ([26] and S. Hanassy, A. Botvinnik, T.F., and B.H., unpublished data). This type of motor program fits the special distribution of control between the ~170 million neurons in the central brain (together with the optic lobes), and the ~330 million neurons in the elaborate and rather autonomous peripheral nervous systems of the arms [4, 5]. For example, the octopus reaching movement has only three adjustable degrees of freedom (df) [1] and can be generated in amputated arms [2], suggesting that central commands are necessary only to activate and scale peripheral motor programs (i.e., motor primitives [2, 27]). It is conceivable that the pushing by elongation is also a motor primitive generated by a predefined motor program that is embedded in the peripheral nervous system of the arms.

The fact that all recruited arms generate similar thrust forces further simplifies the control because the ratio between the forces doesn't need to be considered. This allows a feedforward type of crawling control, where the controlled variable is simply which arms to recruit. Such a simplification has already been demonstrated in the octopus reaching movement, where arms that are extended together have similar velocity profiles,

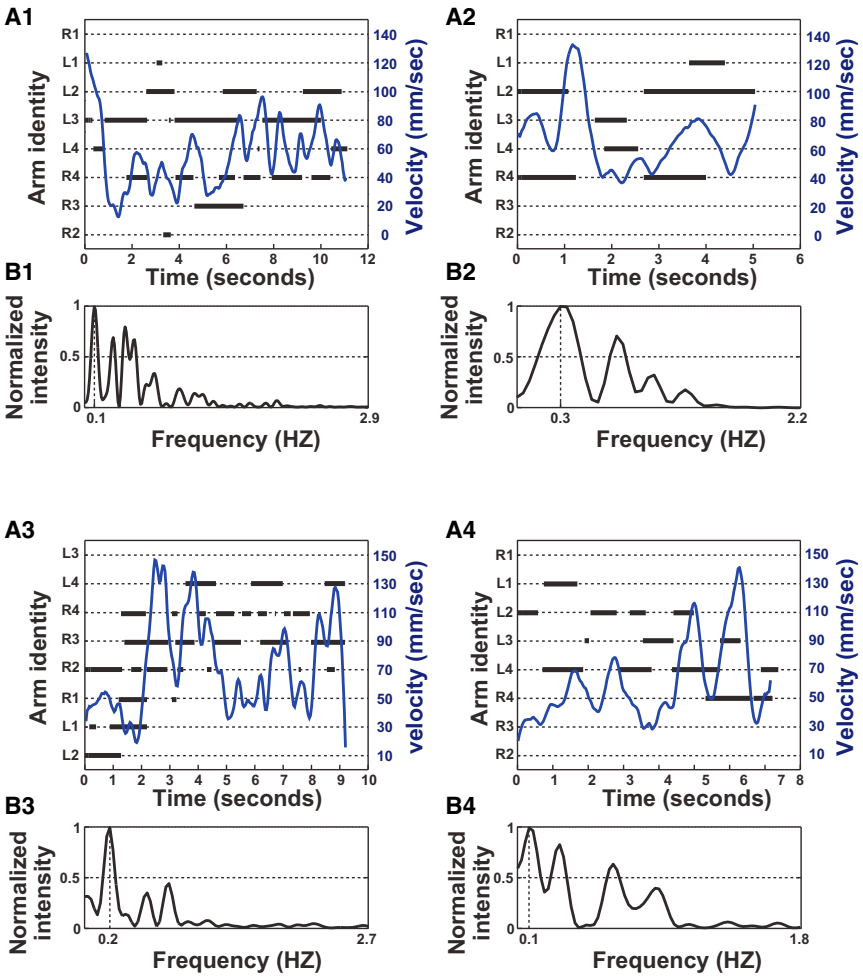


Figure 4. Lack of Stepping Pattern in Octopus Crawling

(A1–A4) Examples of stepping records from different crawling octopuses. Each horizontal dashed line represents one arm (arm identity on the left), with accentuated parts indicating when this arm actively participated in the movement. The instantaneous mouth velocity is superimposed in blue (with respect to the right axis).

(B1–B4) The normalized frequency spectrum extracted from the velocity in the panel above it (shown in A1–A4, respectively) by the fast Fourier transform (FFT). Note the absence of any characteristic frequency peak.

See also Figure S3.

suggesting they are controlled together by a single central motor command [1].

Can the morphology and the biomechanics explain the unusual arm coordination in crawling? It is widely accepted that locomotion patterns, both in vertebrates and invertebrates, are driven by dedicated neural circuits that function as CPGs, which generate a repetitive rhythmical pattern of motor output [16, 17, 28–31]. The fact that we could not find any rhythmical temporal pattern in arm recruitment in octopus crawling raises the intriguing possibility that contrasting locomotion in other animals, the central control of octopus crawling generates a complex temporal pattern that is hard to detect (at least with our techniques) or perhaps does not exist at all. We suggest that a *central command generator* of crawling exists in the motor centers of the octopus brain and directly commands what appears to be ad hoc or moment-to-moment recruitment of octopus arms for instantaneously controlling the direction of crawling, similar to the generation of saccadic eye movements by the superior colliculus [32–34]. The analogy between saccadic eye movements and moment-to-moment control of octopus crawling raises the possibility that the octopus uses body motions like "body saccades" (that were shown to exist in other animals [35]), which result in a controlled motion of the visual image on the octopus retina, a stage that is likely to be important for the processing of visual information [36].

In summary, we believe that our findings provide a striking demonstration of the emergence of special control mechanisms well embodied within the unique morphology and flexibility of the octopus. This is because our results suggest that the special maneuverability of the octopus in crawling emerges from the radially symmetrical distribution of the eight arms around the body and the stereotypical arm pushing movements used for creating the thrust in crawling. These two properties enable the reduction of the complexity involved in the control of motion in a hyper-redundant body to only few controlled variables—the arms identity. Our results also support the embodied-organization concept of adaptive behaviors (in the general sense, not only in octopuses) [4, 5, 37], as they show the marvelously reciprocal interaction between the control system, the physical properties of the body, and the morphology, an interaction that leads to an evolutionary successful adaptation of the emerging behavior to the ecological niche.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Experimental Procedures, three figures, one table, and two movies and can be found with this article online at http://dx.doi.org/10.1016/j.cub.2015.02.064.

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Supplemental Information

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Supplemental Figures

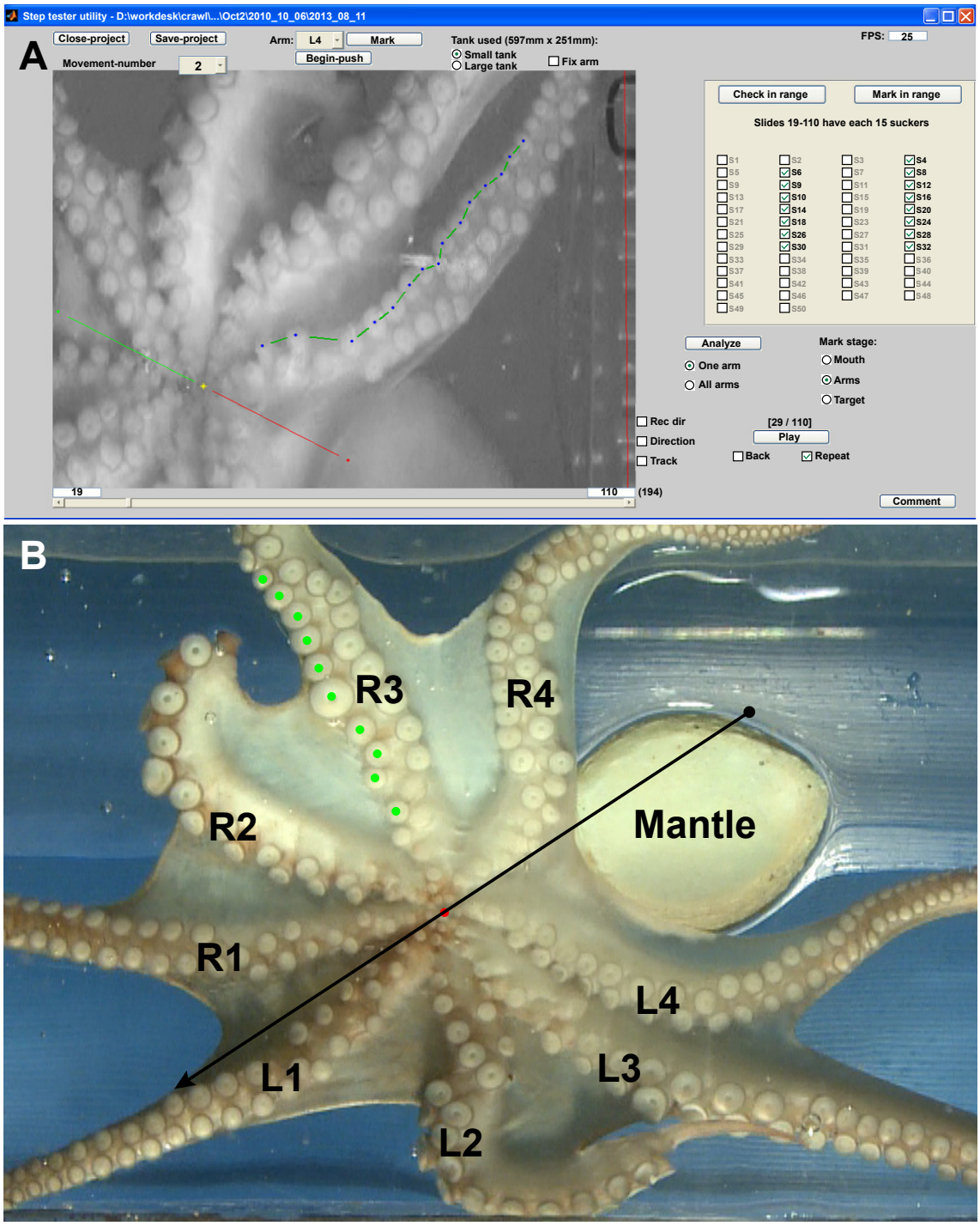


Figure S1. Related to inset in Figure 3. Software for analysing crawling. (**A**) A snapshot of the graphical Matlab software designed for labeling video images and analysing octopus crawling. (**B**) A video image with labeling and with mantle and arms identified. The mouth is marked by the red dot, some suckers' centres by green dots, and the body facing direction is marked by the black arrow. Note the proximity of the first sucker from each arm to the mouth (less than 0.5cm).

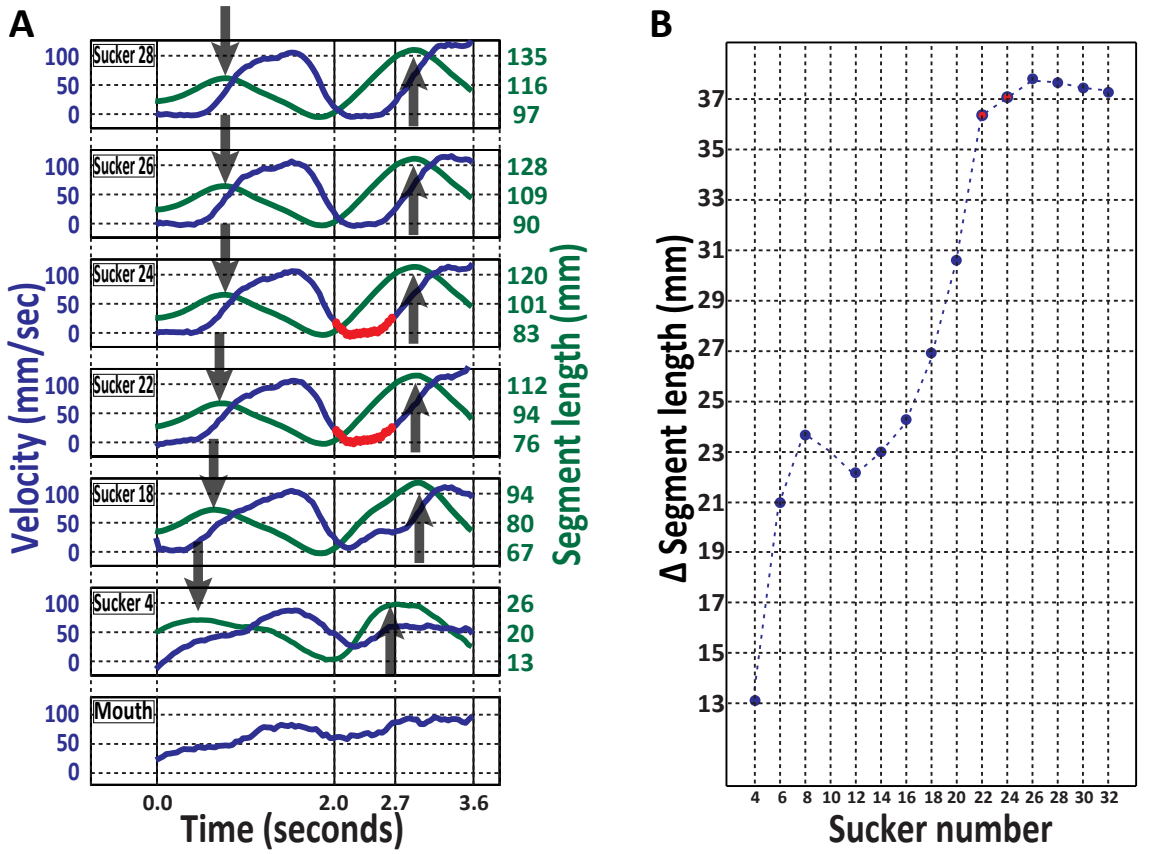


Figure S2. Related to Figure 2. The analysis of a single 3.6 seconds crawling step of one arm. (**A**) Blue plots show the velocities of the mouth (*i.e.*, the body) and of specific suckers (with respect to the left axis). The distance of each sucker from the mouth (segment length) is shown as a green line (with respect to the right axis). The identity of the sucker (counting from the mouth distally) or the mouth is given in the upper left corner of each sub-panel. Red segments on blue lines depict time intervals during which the sucker was attached to the substrate. Downward arrows mark the beginning of the shortening phase and upward arrows mark the end of the elongation phase for each sucker. Vertical lines indicate the beginning and end of the step (between 0.0 and 3.6 seconds) and of the active pushing phase (between 2.0 and 2.7 seconds). The length of the entire arm was 335mm. (**B**) Change in segment length to each sucker shown in A (and to additional suckers) during the active pushing phase (between 2.0 and 2.7 seconds).

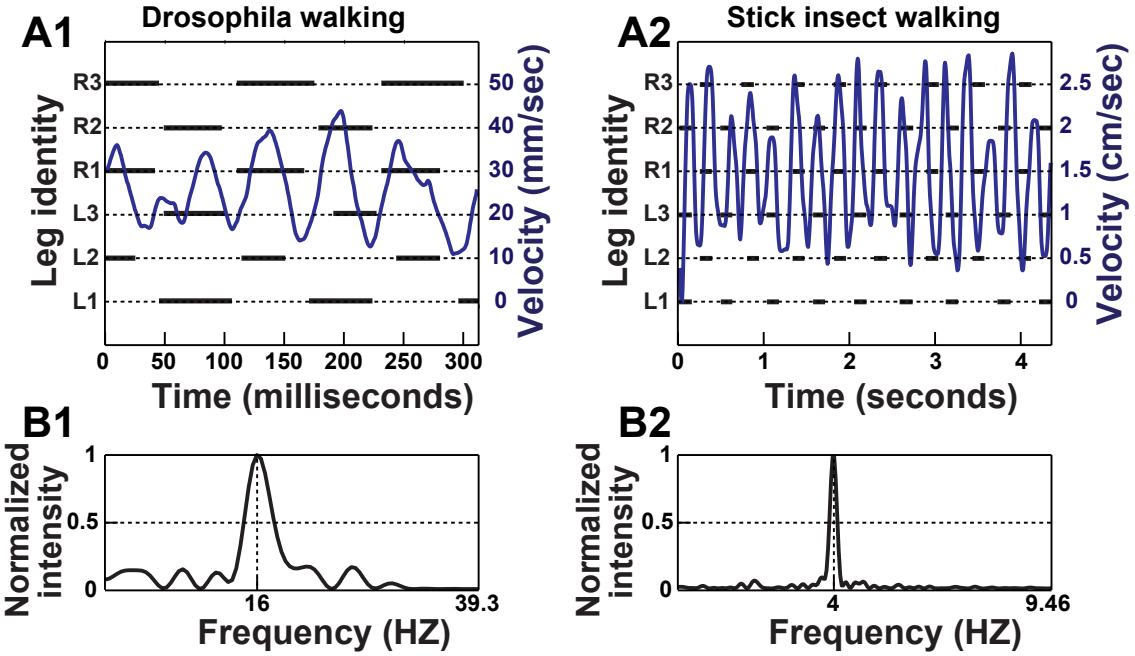


Figure S3. Related to Figure 4. Stepping records and velocities of other animals. (**A1-A2**) Two examples of stepping records of a walking *Drosophila* and a walking stick insect. Each horizontal dashed line represents one leg (leg identity on the left), with accentuated parts indicating when this leg actively participated in the movement. The instantaneous velocity is superimposed in blue (with respect to the right axis). (**B1-B2**) The normalized frequency spectrum extracted from the velocity in the panel above it (shown in panels A1-A2 respectively) by the fast Fourier Transform (FFT). Note the clear characteristic frequency peak of each of the spectra. Right panels, stick insect (based on [S1]); left panels, *Drosophila* (based on [S2]).

L2	**Count=12 χ^2=11.26 *p=0.0013**						
L3	Count=42 χ^2=0.64 p=0.4976	Count=50 χ^2=2.65 p=0.1294					
L4	**Count=5 χ^2=22.5 *p<0.0001**	Count=28 χ^2=0.78 p=0.4463	**Count=84 χ^2=20.18 *p<0.0001**				
R4	**Count=0 χ^2=35 *p<0.0001**	**Count=92 χ^2=25.58 *p<0.0001**	Count=26 χ^2=1.33 p=0.3078	**Count=64 χ^2=8.49 *p=0.0049**			
R3	**Count=9 χ^2=15.36 *p=0.0002**	**Count=0 χ^2=35 *p<0.0001**	**Count=61 χ^2=7.04 *p=0.0107**	Count=43 χ^2=0.82 p=0.431	**Count=74 χ^2=13.95 *p=0.0003**		
R2	**Count=10 χ^2=13.89 *p=0.0003**	**Count=15 χ^2=8 *p=0.0072**	**Count=0 χ^2=35 *p<0.0001**	**Count=93 χ^2=26.28 *p<0.0001**	Count=39 χ^2=0.22 p=0.729	Count=20 χ^2=4.09 p=0.0592	
R1	Count=54 χ^2=4.06 p=0.0564	Count=45 χ^2=1.25 p=0.3125	Count=25 χ^2=1.67 p=0.2435	**Count=0 χ^2=35 *p<0.0001**	Count=25 χ^2=1.67 p=0.2435	Count=27 χ^2=1.03 p=0.3711	Count=37 χ^2=0.06 p=0.8875
	L1	L2	L3	L4	R4	R3	R2

Table S1. Related to Figure 3B. Each cell in the table shows the count of using a specific pair of arms (arm identities are given below the cell on lower row and to the left of the cell in the left column) with the results of the Pearson χ^2 goodness of fit test related to it. The hypothesis for each specific pair claims that the observed (counted) value is uniformly distributed with the average count over all pairs (=35) and is therefore statistically equal to it. Df=1 for all cells and therefore, according to Bonferroni correction, the significance threshold is $p<\frac{0.05}{2}\leq 0.025$**.** Cells in which the test successfully rejected the hypothesis are in bold font and marked with an asterisk.

Supplemental Experimental Procedures

Animals and maintenance

The experiments on nine adult (4 females and 5 males) *Octopus vulgaris* (common octopus) and the analyses were carried out at the Hebrew University of Jerusalem according to the guidelines for the EU Directive 2010/63/EU for cephalopod welfare [S3]. Octopuses are easily acclimatized to captivity and are therefore suitable for research. The animals were bought from fishermen who have been working with our lab for more than 15 years. Each animal was kept alone in a tank which was part of a closed well equilibrated and daily monitored system of synthetic sea water (Aqua Medic) running through biological filters. Each tank was enriched with sand, stones, and green algae. The animals that participated in the experiments for this paper were adults between 120 and 550 grams and the distance between the tip of the mantle to the tip of the head varied between 75 and 200mm. Octopuses size at maturity cannot provide a direct indication of age or condition of health [S4-S7].

Experiment procedure

A single animal was placed in an aquarium with a transparent Perspex floor. The aquarium was filled with SSW to a level of about half the animal's height to prevent it from swimming and to encourage it to move only by crawling, while still enabling it to breathe normally. Note that in nature octopuses routinely leave the water to crawl between rocks on the shore. The experimental aquarium did not contain any sand or rocks. For video recording the animal from underneath, a Sony PMW-EX1R camcorder on a fixed tripod was either placed directly beneath and perpendicular to the tank floor or perpendicular to the image from a mirror aligned at 45° under the tank. The camera focused on the ground plane on which the animal crawled and was adjusted to record the whole tank floor. Recordings were made at high definition resolution (1920x1080) and at a frame rate of 25 frames per second, with a shutter speed of 1/300 to avoid blurring. For calibration (explained below), an object of known dimensions was placed on the video-recorded plane for at least some of the recording session (in most cases the object was part of the tank floor itself).

Octopuses are intelligent animals as is overtly demonstrated by their clear acclimatization process to the captivity. During the first 1-2 weeks in the lab they start to go out of their

den (an upside down clay pot) and to move around in their aquarium. Then due to their well known natural curiosity [S8], they continuously move around in the tank, seemingly interested in what's happening around them. This behavior can be facilitated by movements of the experimenters or caretakers or by moving a finger in front of them as the octopuses cannot resist attacking moving objects (if not too big). Therefore in our experiment we used well acclimatized animals (at least for two weeks). These animals were crawling spontaneously and when stopped it was easy, in most cases, to encourage them to crawl again by simply moving around or teasing them with a moving finger outside the experimental tank. In several cases we introduce food to the tank in order to encourage them to crawl again towards a target. In some experiments, obstacles were placed on the tank floor to encourage the animal to change crawling direction. In the sessions that were video recorded through a mirror, the camera also captured the animal directly from the side to evaluate elevation of the arms during crawling. The animal was recorded for ten minutes or until it stopped moving around (whichever ended first), and then returned, in a gentle way to minimize stressing the animal, to its home aquarium. Each animal participated in one session per day at the most.

Data acquisition and analysis

All data were derived from video clips recorded during the experiments. Video movies were cut into single images using the commercial software Adobe Premiere Pro CS4, converted into black-and-white images and stored in directories. Only those parts of the videos in which the octopus crawled continuously were used for creating the images.

Labeling video images

Several custom-made Matlab programs with suitable Graphical User Interface (GUI) were designed and used for labeling and analysing the data. The video images were loaded into the GUI and points of interest in the images were labeled and stored as additional data. Figure S1 shows, as a general example, a snapshot from one of the Matlab programs and a labeled video image. Although the mouth is not the animal's centre of mass, it is the point from which all the arms emerge, and therefore we labeled the mouth and regarded it as the centre of the animal's body in terms of position and motion (Figure S1B). Some of the suckers were labeled as markers of specific points on the arms across images (green dots in Figure S1B); the suckers are easy to see on the images, their centres are easy to identify, and they lay at fixed locations on the arm. We

labeled only few suckers of interest out of the about 300 suckers that are organized in two dense rows along the ventral (oral) side of each arm. All velocities are with respect to the ground and not with respect to the body. We also added additional information stating whether or not the sucker was attached to the substrate in the labeled image for the analysis of the arms that were actively participating in crawling. The octopus has a well defined front and back. The soft and very flexible body of the animal can be misleading and conceal the accurate body direction, and the vector between the four left arms and the four right arms is the only line that always reliably merges with the true and accurate body orientation of the animal. The mantle, which emerges from the back side of the body was used for an initial approximation of the body facing direction, but the exact direction was labeled as the orientation of that vector (black arrow in Figure S1B).

Calibration

Calibration was required to convert the distance between two points in the video frame (in numbers of horizontal and vertical pixels) to real distances. To calibrate each video clip, two one-dimensional objects with known lengths were placed in the scene and the edges of the objects were labeled. Calculation of distance was based on trivial geometry by creating two equations with two unknown variables. One side of the equation gave the length of the object using the Pythagorean Theorem and the other side gave the known length of the object. To obtain the horizontal (X) and vertical (Y) dimensions of an item represented by one pixel, we take an object of known length L, and the horizontal and vertical number of pixels between its edges H_n and V_n and get $(H_n \bullet X)^2 + (V_n \bullet Y)^2 = L^2$.

Because there are two calibration objects we have two equations, each with the same two unknowns, X and Y, so the equations can be solved to find X and Y. For obvious reasons the objects cannot be positioned in parallel, and an angle closer to perpendicular will yield more accurate results. Therefore, we positioned the objects at an angle close to 90° relative to each other.

Kinematic parameters calculations

After calibration and labeling all the video images of a continuous clip, the distances between labeled points on an image could be measured. Since the camcorder was stationary during the entire clip, the distance travelled by a point between images could be measured as well, and in addition the camera's frame rate was used to calculate velocities.

Significance of arm usage

Our results show that octopuses prefer using specific arms and arm-pairs over others (see main text). All the differences were found to be statistically significant, as the χ^2 Pearson goodness of fit test rejected all hypotheses claiming that the observed values are statistically equal (uniformly distributed), with the following:

1. For the hypothesis that all the eight arms usage frequency was uniformly distributed (Figure 3A): df=7, χ^2=163.478, p<0.0001 (according to Bonferroni correction the significance threshold is $p<\frac{0.05}{8}\leq 0.00625$).

2. For the hypothesis that the extra usage of hind arms (count=1516) vs. front arms (count=990) was uniformly distributed (Figure 3A): df=1, χ^2=110.41, p<0.0001 (according to Bonferroni correction the significance threshold is $p<\frac{0.05}{2}\leq 0.025$).

3. For the hypothesis that the extra usage of the third arms (L3 and R3 – count=810) vs. the forth arms (L4 and R4 – count=706) was uniformly distributed (Figure 3A): df=1, χ^2=7.13, p=0.0082 (according to Bonferroni correction the significance threshold is $p<\frac{0.05}{2}\leq 0.025$).

4. For the hypothesis that the usage frequency of all 28 pairs of arms was uniformly distributed (Figure 3B): df=27, χ^2=620.114, p<0.0001 (according to Bonferroni correction the significance threshold is $p<\frac{0.05}{28}\leq 0.00179$).

5. During crawling some pairs of arms were recruited together more often than other pairs (see also Figure 3B). The count for each pair of arms is given in Table S1. The average over the counts of all pairs was 35. Hypothesizing that the count for each specific pair of arms was statistically equal to the average (originated from a uniform distribution) was tested for each pair separately and the results are given in Table S1. The pairs of arms that were found to be used statistically more often than the average were L3+L4, L4+R4, L2+R4, L3+R3, L4+R2, and R4+R3, and the pairs that were found to be used statistically less often than the average were L1+L2, L4+L1, R3+L1, R2+L2, and R2+L1. Note that the pairs L1+R4, L2+R3, L3+R2, and L4+R1 were never recruited together likely because these are pairs of arms that push in opposite directions, these of course were also found to be used statistically less often than the average

Arm behaviour in crawling

To analyse arm behaviour in crawling we followed some suckers that were labeled across consecutive video images. The suckers on the arms are good labels because their round shape makes them easy to detect and, they are organized in a fixed pattern along the arm. The first sucker of each arm is part of a small, easy to identify, circle of eight suckers with a fixed radius of about 0.5cm around the mouth (Figure S1B). We used the position and motion of the mouth for representing *de facto* the position and motion of the animal body with respect to the ground. The velocity (with respect to external coordinates) was measured directly as the derivative of the position between two consecutive time frames. Time intervals during which the suckers were attached to the substrate are also shown (red).

1) Pushing by elongation behaviour

To create the thrust for pushing the body in crawling, octopus arms used a stereotypical shortening-elongating behaviour that consisted of several stages. Figure S2A presents an example analysis of one stereotypical pushing step by showing the velocities of the body (*i.e.,* the mouth) and of the specific chosen suckers (blue) along with the length of the arm segment between the mouth and each of these suckers (green). First, the arm shortened, bringing the suckers closer to the mouth (for each sucker in Figure S2A, the time interval between the arrow pointing downwards and the vertical line pointing at 2.0 seconds). Then, a group of adjacent suckers attached to the substrate as an 'anchor' (suckers 22 and 24 in Figure S2A). While this anchor was attached to the substrate, the part of the arm between the mouth and the anchor elongated, creating thrust between the body and the anchor, thus pushing the body to move. In the analysed example, suckers 22 and 24 were part of the anchor and they were attached to the substrate between 2.0 and 2.7 seconds. The more proximal suckers (between 4 and 18), located on the arm between anchor and mouth, moved (with respect to the ground) during the elongation. The more distal suckers (26 and 28) moved passively with virtually the same velocity profile as the anchoring suckers, showing that only the part of the arm proximal to the anchor elongated. Changes in segment length of suckers between the anchor and the mouth (Figure S2B) show there was no consistent temporal order in which the proximal arm segment elongated but this might have been obscured by the movement of the free suckers on the sucker stalk.

2) Arm behaviour for rotating the body

Here we provide only a qualitative description of the two types of arm behaviour for rotating the body. In both strategies, a lateral bend was formed in the active arm and then some suckers distal to the bend attached to the substrate to form an anchor. Next, the part of the arm proximal to the anchor stiffened, straightening the bend without any clear change in the length of the arm segment. Following this stage, one of two strategies were used to generate rotation; either the angle of the bend directly increased, producing radial thrust that pushed the body forward and rotated it at the same time, or the anchoring suckers were sequentially released from the substrate from proximal to distal while the proximal segment straightened, thereby causing a rotation of the body.

3) Randomness in stepping duration

To search for a temporal pattern in arm recruitment during octopus crawling, regardless of the spatial pattern of distribution they originated from, we collected the step durations of all the arms from 12 continuous octopuses crawling movements. Octopuses use all arms in the same way in crawling, so we did not separate data from different arms. In order to eliminate the possibility that randomness is due to velocity differences, we multiplied each step duration with the average velocity of the movement (because they have negative linear ratio). The extended distribution free Wald–Wolfowitz runs-test [S9] (described in Bradley [S10]) on 84 steps could not reject the hypothesis that the numbers were random (number of runs: 48 $z=-1.0348$, $p=0.3008$). Similar tests on the step durations of a walking stick-insect and *Drosophila* (based on data from Graham [S1] and Mendes, et al. [S2] respectively) rejected randomness (stick-insect: 5 continuous movements, 128 steps, number of runs=26, $z=-6.7209$, $p<0.0001$. *Drosophila*: 2 continuous movements, 78 steps, number of runs=6 $z=7.4020$, $p<0.0001$), supporting our technique and therefore our general conclusion that arm recruitment in octopus crawling lacks a clear temporal pattern.

Supplemental References

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